ECE 329 Introduction to Electromagnetic Fields Spring 18

University of Illinois He, Goddard, Ilie

Exam 3

Wednesday, April 18, 2018 — 7:15-8:30 PM

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the boxes below.

Name:	Solutions
Section:	12 Noon 1 PM 2 PM

If you want your exam returned to a **different** section, write it here:

Special Return to Section:	12 Noon 1 PM 2 PM

This is a closed book exam and calculators/electronic devices are not allowed. You are allowed THREE double sided 3"x5" index card of notes. Please show all your work and make sure to include your reasoning for each answer. All answers should include **units** wherever appropriate. The exam is double sided. You may use the back of the exam as scratch paper.

Problem 1 (30 points)	
Problem 2 (20 points)	
Problem 3 (30 points)	
Problem 4 (20 points)	
TOTAL (100 points)	

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Because calculators are not allowed, please use the following expressions for physical constants:

$$c = 3 \times 10^8 \left[\frac{m}{s}\right]$$

$$\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \left[\frac{F}{m}\right]$$

$$\mu_0 = 4\pi \times 10^{-7} \left[\frac{H}{m}\right]$$

$$\eta_0 = 120\pi \ [\Omega]$$

and leave your answers in the form of fractions rather than decimals.

A Table for Your Reference

23 Imperfect dielectrics, good conductors

	Condition	β	α	$\lvert\eta\rvert$	τ	$\lambda = \frac{2\pi}{\beta}$	$\delta = \frac{1}{\alpha}$
Perfect dielectric	$\sigma = 0$	$\omega\sqrt{\epsilon\mu}$	0	$\sqrt{\frac{\mu}{\epsilon}}$	0	$\frac{2\pi}{\omega\sqrt{\epsilon\mu}}$	∞
Imperfect dielectric	$\frac{\sigma}{\omega\epsilon} \ll 1$	$\sim \omega\sqrt{\epsilon\mu}$	$\beta\frac{1}{2}\frac{\sigma}{\omega\epsilon} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$	$\sim \sqrt{\frac{\mu}{\epsilon}}$	$\sim \frac{\sigma}{2\omega\epsilon}$	$\sim \frac{2\pi}{\omega\sqrt{\epsilon\mu}}$	$\frac{2}{\sigma}\sqrt{\frac{\epsilon}{\mu}}$
Good conductor	$\frac{\sigma}{\omega\epsilon} \gg 1$	$\sim \sqrt{\pi f \mu\sigma}$	$\sim \sqrt{\pi f \mu\sigma}$	$\sqrt{\frac{\omega\mu}{\sigma}}$	45°	$\sim \frac{2\pi}{\sqrt{\pi f\mu\sigma}}$	$\sim \frac{1}{\sqrt{\pi f\mu\sigma}}$
Perfect conductor	$\sigma = \infty$	∞	∞	0	-	0	0

- The table above summarizes TEM wave parameters in homogeneous conducting media where the propagation velocity

$$v_p = \frac{\omega}{\beta}$$

x-polarized phasor

$$\tilde{\mathbf{E}} = \hat{x} E_o e^{\mp\alpha z} e^{\mp j\beta z}$$

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1. (30 points) Multiple choice questions (3 points each). Circle only **ONE** letter next to the **ONE** correct statement. Note that most of these questions require some calculation to get the right answer, but you do not need to show your work to get credit. No partial credit is given on the multiple choice questions.

(i) What statement applies to a wave propagating inside a good conductor:

(a) The attenation constant is approximately zero

(b) The phase constant is a linear function of frequency

(c) The wave amplitude decreases a small amount as the wave travels a distance of $1\,\lambda$

(d) The electric and magnetic fields are 45 degrees out of phase

(e) None of the above

(ii) Telegrapher's equations for voltage and current along a transmision line:
I. can be derived directly from Maxwell's equations for electric and magnetic fields
II. result in d'Alembert solutions of the wave equation (i.e., $f(t-\frac{z}{v})$ and $g(t+\frac{z}{v})$)
III. equivalently describe a distributed circuit consisting of capacitors in series interweaved with inductors shunting to ground ← no, inductors in series and capacitors shunting to ground

(a) I only

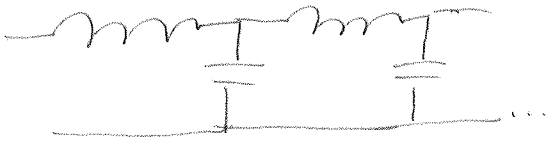
(b) II only

(c) III only

(d) I and II only

(e) I, II, and III

(f) None of the above



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(iii) Consider a plane wave propagating in free space with electric field given by:

$$\tilde{\mathbf{E}} = 10e^{-j\beta z}\hat{x} + Be^{j\phi}e^{-j\beta z}\hat{y}\ \text{V/m}$$

where we assume that B and ϕ are **real non-negative numbers** : $B \geq 0$ and $0 \leq \phi < 2\pi$.
I. The wave will be linearly polarized if $\phi = 0$ for any value of B.
II. The wave will be linearly polarized if $\phi = \frac{\pi}{2}$ for any value of B.
III. The wave will be linearly polarized if $B = 0$ for any value of ϕ.

(a) I only
(b) II only
(c) III only
(d) I and III only
(e) I, II, and III
(f) None of the above

I. if $\phi = 0 \Rightarrow \tilde{E} = e^{-j\beta z}(10\hat{x} + B\hat{y})$ is linear

II. No. $\phi = \pi/2$ will lead to elliptical polarization for different values of B (circular if $B=10$, linear if $B=0$, elliptical otherwise)

III. if $B=0 \Rightarrow \tilde{E} = 10e^{-j\beta z}\hat{x}$ is linear

(iv) Consider the same plane wave as in (iii) above.
I. The wave will be right circularly polarized if $\phi = 0$ and $B = 10$.
II. The wave will be right circularly polarized if $\phi = \frac{\pi}{2}$ and $B = 10$.
III. The wave will be right circularly polarized if $\phi = \frac{3\pi}{2}$ and $B = 10$.

(a) I only
(b) II only
(c) III only
(d) I and III only
(e) I, II, and III
(f) None of the above

$B=10$ and $\phi = \frac{\pi}{2}$ or $\frac{3\pi}{2}$ for circular

Wave is moving in $+\hat{z}$

RHCP if $\mathrm{Re}[\hat{\tilde{E}}] \times \mathrm{Im}[-\hat{\tilde{E}}] =$ propagation direction $= +\hat{z}$

$\mathrm{Re}[\hat{\tilde{E}}] = 10\hat{x}$

$\mathrm{Im}[-\hat{\tilde{E}}] = \begin{cases} \mathrm{Im}[-jB\hat{y}] \text{ for } \phi = \pi/2 & = -\hat{y} \\ \mathrm{Im}[jB\hat{y}] \text{ for } \phi = \frac{3\pi}{2} & = \hat{y} \end{cases}$

$\hat{x} \times \hat{y} = \hat{z}$ so it has to be $\phi = \frac{3\pi}{2}$

$e^{j\pi/2} = j$
$e^{j\frac{3\pi}{2}} = -j$

Alternatively, we can write out the fields

$E_x = 10\cos(\omega t - \beta z)$

$E_y = \mathrm{Re}[Be^{j(\omega t - \beta z + \phi)}] = B\cos(\omega t - \beta z + \phi) = \begin{cases} -B\sin(\omega t - \beta z) & \text{if } \phi = \pi/2 \\ B\sin(\omega t - \beta z) & \text{if } \phi = \frac{3\pi}{2} \end{cases}$

Ahead field is E_x
Behind field is E_y
and Ahead × Behind = prop. direction
so this is RHCP

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(v) Be sure to consider polarity in this question: the characteristic impedance of a transmission line Z_0 is equal to the ratio of the voltage to the current for the:
I. incident wave
II. reflected wave
III. sum of incident and reflected waves

$\frac{V^+}{I^+} = Z_0 \qquad \frac{V^-}{I^-} = -Z_0$

$Z(d) = \frac{V}{I} = \frac{V^+ + V^-}{I^+ + I^-} = Z_0 \frac{1 + \frac{V^-}{V^+}}{1 - \frac{V^-}{V^+}} = Z_0 \frac{1 + \Gamma(d)}{1 - \Gamma(d)}$

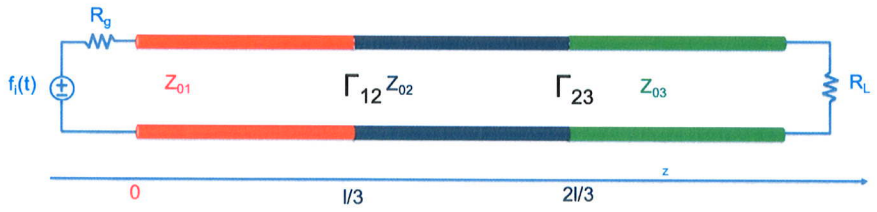
(a) I only
(b) II only
(c) III only
(d) I and II only
(e) I, II, and III
(f) None of the above

(vi) For a single transmission line (TL) terminated in an arbitrary load, the generalized reflection coefficient $\Gamma(d)$:
I. is equal to the ratio of the voltage phasor of the reflected wave to that of the incident wave at every location along the TL yes
II. has constant magnitude along the TL yes
III. has constant phase along the TL no

$\Gamma(d) = \frac{\tilde{V}^-(d)}{\tilde{V}^+(d)} = \Gamma_L e^{-2j\beta d}$

(a) I only
(b) II only
(c) III only
(d) I and II only
(e) I, II, and III
(f) None of the above

(vii) Given the setup in the picture below, what is the minimum necessary condition for there to be no wave reflected back to the source:



(a) $\Gamma_{12} = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} = 0$
(b) $\Gamma_{23} = \frac{Z_{03} - Z_{02}}{Z_{03} + Z_{02}} = 0$
(c) $\Gamma_{12} = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} = 0$ and $\Gamma_{23} = \frac{Z_{03} - Z_{02}}{Z_{03} + Z_{02}} = 0$
(d) None of the above because more conditions are needed also need $R_L = Z_{03}$

(Note that it is ok for $Z_{01} \neq R_g$)

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(viii) Consider a transmission line (TL) shorted at the load end. The characteristic impedence of the TL is Z_0. Assume V^+ has a constant value $\neq 0$. Then, the current phasor along the TL is:

(a) $\tilde{I}(d) = \frac{V^+}{Z_0} e^{j\beta d}$

(b) $\tilde{I}(d) = \frac{V^+}{Z_0} e^{-j\beta d}$

(c) $\tilde{I}(d) = \frac{V^+}{Z_0} e^{j\beta d} + \frac{V^+}{Z_0} e^{-j\beta d}$

(d) $\tilde{I}(d) = \frac{V^+}{Z_0} e^{j\beta d} - \frac{V^+}{Z_0} e^{-j\beta d}$

(e) None of the above

Shorted $\Rightarrow Z_L = 0 \Rightarrow \Gamma_L = \frac{0 - Z_0}{0 + Z_0} = -1$

$\tilde{I}(d) = \frac{V^+}{Z_0} e^{j\beta d} - \frac{V^-}{Z_0} e^{-j\beta d}$

$= \frac{V^+}{Z_0} e^{j\beta d} + \frac{V^+}{Z_0} e^{-j\beta d}$ because $V^- = \Gamma_L V^+$

(ix) Consider a TL of length l that is shorted at the load end. If the lowest series resonance frequency is ω_0, then the input line impedence $Z_{in} = Z(l)$ at frequency $\omega = 2\omega_0$ is:

(a) $Z_{in} = Z(l) = 0$

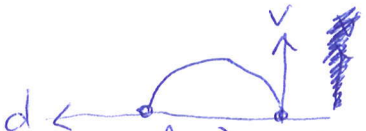
(b) $Z_{in} = Z(l) = \infty$

(c) $Z_{in} = Z(l)$ is inductive

(d) $Z_{in} = Z(l)$ is capacitive

(e) None of the above

Series resonance means $Z_{in} = 0$



$l = \frac{\lambda_0}{2}$ at the fundamental frequency because it is shorted at both ends

if $\omega = 2\omega_0 \Rightarrow \lambda = \frac{\lambda_0}{2}$ (assuming the velocity is constant)

$\Rightarrow l = \lambda$



also shorted

(x) Consider a TL of length l that is shorted at the load end. If the lowest series resonance frequency is ω_0, then the input line impedence $Z_{in} = Z(l)$ at frequency $\omega = \frac{1}{2}\omega_0$ is:

(a) $Z_{in} = Z(l) = 0$

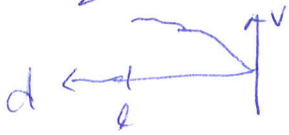
(b) $Z_{in} = Z(l) = \infty$

(c) $Z_{in} = Z(l)$ is inductive

(d) $Z_{in} = Z(l)$ is capacitive

(e) None of the above

if $\omega = \frac{1}{2}\omega_0 \Rightarrow \lambda = 2\lambda_0$ and so $l = \frac{\lambda}{4}$



open

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2. (20 points) The region $y < 0$ is free space whereas the region $y > 0$ is a perfect dielectric with $\mu = \mu_0$. A wave propagating from the region $y < 0$ in the $+y$ direction is incident on the interface. The electric field phasor for the **transmitted** wave is given by $\tilde{E}_t = 10e^{-j12y}\hat{z}$ [V/m] for $\omega = 3 \times 10^9$ [rad/s]. A useful identity for this problem is $\eta\gamma = j\omega\mu$.

a) (4 pts) What is the propagation constant $\gamma_{free\ space}$ of the waves in the free space region ($y < 0$)?

Alternatively, you can use $\eta_0 = 120\ \pi$ and find $\gamma = j\omega\mu/\eta_0$

Free space $\Rightarrow \alpha = 0$

$c = v = \frac{\omega}{\beta} \Rightarrow \beta = \frac{\omega}{c} = \frac{3 \cdot 10^9}{3 \cdot 10^8} = 10\ m^{-1}$

$\gamma = \alpha + j\beta$

Your Answer (include appropriate units):

$\gamma_{free\ space}$: 10 j rad/m

b) (2 pts) What is γ_t, the propagation constant of the transmitted wave in the region ($y > 0$)?

$\tilde{E}_t = 10\, e^{-\gamma_t y}\, \hat{z}$ for wave moving in $+\hat{y}$ direction

$\Rightarrow \gamma_t = 12j$

Your Answer (include appropriate units):

γ_t: 12 j rad/m

c) (4 pts) What is η_t, the intrinsic impedance in the transmitted region ($y > 0$)?

$\eta_t = \frac{j\omega\mu}{\gamma_t} = \frac{j \cdot 3 \cdot 10^9 \cdot 4\pi \cdot 10^{-7}}{12j} = \pi \cdot 100$

Your Answer (include appropriate units):

η_t: 100π Ω

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d) (4 pts) What are the electric field reflection and transmission coefficents Γ and τ for the wave from the region $y < 0$ as it impinges on the interface at $y = 0$?

Note $\eta_1 = 120\pi\ \Omega$ for free space

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{100\pi - 120\pi}{100\pi + 120\pi} = \frac{-20}{220} = -\frac{1}{11}$$

$$\tau = 1 + \Gamma = \frac{10}{11}$$

Your Answer (include appropriate units):	
Γ:	$-\frac{1}{11}$ (no units)
τ:	$\frac{10}{11}$ (no units)

e) (6 pts) Find the **incident** and **reflected** electric field phasors for the region $y < 0$.

$$E_t(y=0) = \tau E_{in}(y=0)$$

$$\Rightarrow 10 = \tau E_0$$

$$\Rightarrow E_0 = \frac{10}{\tau} = \frac{10}{10/11} = 11$$

$E_{in} = E_0 e^{-j10y} \hat{z}$ where E_0 could be complex. This would be the most general possible incident wave

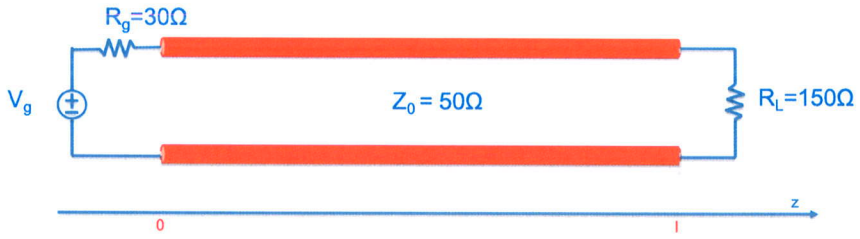
$$E_r = \Gamma E_{in} \text{ at } y = 0$$

$$\Rightarrow E_r = -\frac{1}{11} \cdot 11\, e^{+j10y} \hat{z}$$ because wave moves in $-\hat{y}$ direction

Your Answer (include appropriate units and vector direction):	
$\tilde{E}_{incident}$:	$11\, e^{-j10y}\, \hat{z}$ V/m
$\tilde{E}_{reflected}$:	$-1\, e^{+j10y}\, \hat{z}$ V/m

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3. (30 points) Consider a transmission line with characteristic impedance $Z_0 = 50\Omega$, length $l = 15$cm, and propagation speed $v_p = \frac{2}{3}c$. A voltage source with internal resistance $R_g = 30\Omega$ is connected at the $z = 0$ end, while a load resistance $R_L = 150\Omega$ terminates the line at $z = l$.



a) (6 pts) Calculate the injection coefficient, the source reflection coefficient, and the load reflection coefficient.

$\tau_g = \frac{50}{50+30} = \frac{5}{8}$

$\Gamma_s = \frac{30-50}{30+50} = \frac{-1}{4}$

$\Gamma_L = \frac{150-50}{150+50} = \frac{1}{2}$

Your Answer:
(i) = injection coefficent, τ_g: 5/8
(ii) = source reflection coefficient, Γ_s: -1/4
(iii) = load reflection coefficient, Γ_L: 1/2

$V = \frac{2}{3}c = 200\frac{m}{\mu s}$

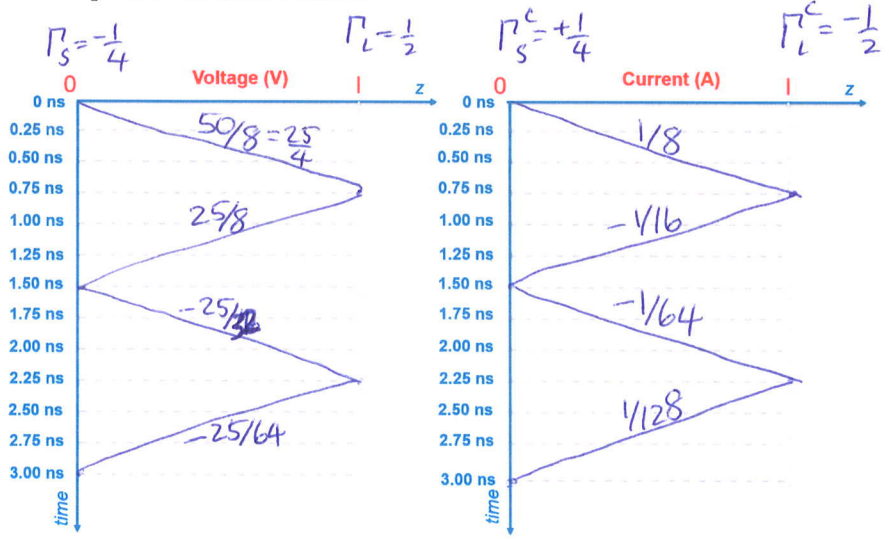
b) (12 pts) Construct the bounce diagrams for voltage and for current, given $V_g = 10u(t)\,V$. Label the wave amplitude on each bounce.

$T = \frac{l}{v_p} = \frac{0.15m}{200m/\mu s}$

$= \frac{0.15m}{0.2\,m/ns}$

$= 0.75ns$

$\tau_g \cdot 10 = \frac{5}{8} \cdot 10 = \frac{50}{8}$ or $\frac{25}{4}$



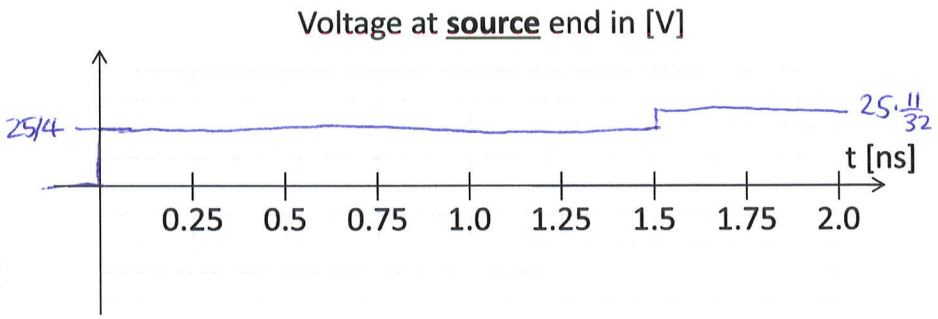
reflection coefficient for current is negative that of voltage

First wave has $I^+ = \frac{V^+}{Z_0} = \frac{50/8}{50} = \frac{1}{8}A$

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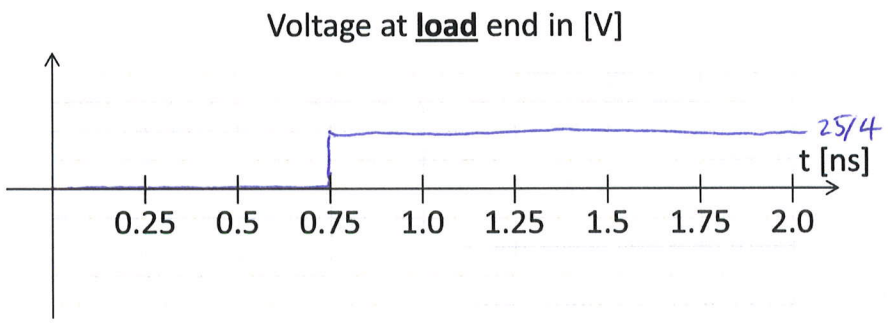
c) (6 pts) Plot the voltage at the source end from t = 0 to t = 2.0 ns. Add numerical values for the vertical axis.

initially 25/4
at time T, it becomes
$\frac{25}{4} + \frac{25}{8} - \frac{25}{32}$
$= 25\,\frac{8+4-1}{32}$
$= 25 \cdot \frac{11}{32}$ which is $25 \cdot \frac{3}{32}$ bigger



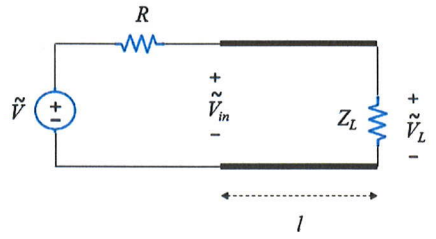
d) (6 pts) Plot the voltage at the load end from t = 0 to t = 2.0 ns. Add numerical values for the vertical axis.

nothing till t=T
then
only the
first wave
reaches
during the
given time
interval



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4. (20 points) Consider a TL segment of propagation velocity $v = 200\,\frac{m}{\mu s}$, characteristic impedance $Z_o = 50\Omega$, and length $l = 50$ m. The transmission line is terminated by a load Z_L. As shown in figure below, the segment is connected *in series* with a resistor $R = 50\Omega$ and sinusoidal voltage source with voltage phasor $\tilde{V} = 2\angle 0$ V.



a) (10 pts) If $Z_L = 0$, and it is found that the current is zero through the resistor R, find all the possible wavelengths λ for the system.

$I_{in} = 0 \Rightarrow Z_{in} = \infty$

So open at input and shorted at load $\Rightarrow l = \text{odd integer} \cdot \frac{\lambda}{4}$

$l = (2n+1)\frac{\lambda}{4} \Rightarrow \lambda = \frac{4l}{2n+1}$ $n = 0, 1, 2, \ldots$ $l = 50m$ so $\lambda = \frac{200m}{2n+1}$

Your Answer (include appropriate units):

$\lambda =$ $\frac{200}{2n+1}$ [m] for $n = 0, 1, 2, \ldots$

b) (10 pts) If instead $Z_L = 100\Omega$, and the voltage source is at frequency such that $l = \frac{\lambda}{4}$, find the impedence Z_{in} and the voltage phaser $\tilde{V}_{in}$ at the input of the transmission line. (Note: the current through the resistor has not been specified for this sub-problem.)

For this quarter wave transformer, $y_{in} = z_L$

$\frac{Z_o}{Z_{in}} = y_{in} = z_L = \frac{Z_L}{Z_o} \Rightarrow Z_{in} = \frac{Z_o^2}{Z_L} = \frac{50 \cdot 50}{100} = 25\,\Omega$

$V_{in} = \frac{Z_{in}}{Z_{in}+R}\tilde{V} = \frac{25}{75}\tilde{V} = \frac{1}{3} \cdot 2\angle 0$

Your Answer (include appropriate units):

$Z_{in} =$ $25\,\Omega$

$\tilde{V}_{in} =$ $\frac{2}{3}\angle 0$ V

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